

Ryan Koh

Boulder, CO | ryankoh.usa@gmail.com | +1 (720) 305-7235 | www.linkedin.com/in/rryankohh

PROFESSIONAL SUMMARY

Mechanical Engineering student at the University of Colorado Boulder with a Computer Science minor and hands-on experience in autonomous systems deployment, robotics field operations, and sensor integration. Proficient in hardware-software troubleshooting across LiDAR and camera sensor arrays, with client-facing experience in aerospace and autonomous systems environments. Seeking a robotics or autonomous systems engineering role where I can apply systems testing, field operations, and cross-functional collaboration skills.

EDUCATION

University of Colorado Boulder

BS in Mechanical Engineering, Computer Science Minor. GPA: 3.68.

May 2027

Relevant Coursework: Calculus I-III, Discrete Math, Data Structures, Algorithms, Computer Systems, Physics I-III, Materials Science, Statics, Differential Equations and Linear Algebra, CAD, Mechanics of Solids, Dynamics, Thermodynamics, Fluid Dynamics, Component Design, Computational Methods, Heat Transfer, System Dynamics, Circuits, Manufacturing, Data Analysis, Graduate Intro to Robotics

Singapore American School

Singapore

June 2023

EXPERIENCE

Robotics Field Operator / Test Engineer

June 2026 – Present

Pattern Labs

Erie, CO, USA

- Deploy and chaperone an autonomous baggage transport robot for commercial flights at Denver International Airport, supporting a \$5 million partnership contract with United Airlines.
- Troubleshoot hardware-software integration issues by diagnosing, retrofitting, and replacing components within a complex LiDAR camera and sensor array, while maintaining accurate documentation and version control.
- Validate new software releases by tracking, organizing, and analyzing operational data to evaluate autonomous robot performance.
- Communicate sensor and software design flaws to the engineering team to optimize autonomous robot performance.
- Collaborate with interdisciplinary teams of technicians, electrical engineers, and software engineers to develop test plans, safety compliance protocols, and performance benchmarks.

Engineer & Research Intern

May 2025 – July 2025

Jebsen & Jessen Technology and Industrial Services

Singapore

- Developed and tested an automated engine inspection system utilizing a drone and photogrammetry software for non-destructive testing of aircraft fuselage and engines.
- Researched and interpreted technical documentation of robotics and drone systems with computer vision capabilities to scope a new hardware and systems project.
- Engaged with clients including SAESL and Pratt & Whitney to define non-destructive testing inspection parameters and system requirements.
- Coordinated enterprise-grade hardware and software acquisition with suppliers within a defined project budget.
- Produced technical documentation and budget reports for the automated aircraft engine inspection system.

Racing Team Member

August 2025 – Present

Formula Society of Automotive Engineers

Boulder, CO, USA

- Design, manufacture, and test a formula-style race car for the annual Formula SAE competition against over 100 university teams nationwide.
- Apply computational fluid dynamics to analyze downforce and optimize aerodynamic components against target requirements including weight, drag, and acceleration.
- Conduct composite material testing using a universal testing machine, gathering stress-strain data to inform design decisions.

SKILLS

Software and Coding: C++, Python, MATLAB, Java, C#, Assembly, GitHub, VS Code, Unity, SimVascular, 3D Slicer, Office 365, Google Suite, Adobe Suite

Robotics and Systems: LiDAR Sensor Integration, Autonomous Systems Testing, Hardware-Software Troubleshooting, Computer Vision, Arduino, Circuits, Soldering, Technical Documentation

Computer-Aided Design: SolidWorks (CSWA), Onshape, Autodesk Fusion 360, Design Drawing, 3D Printing

Fabrication and Testing: Composite Material Testing, Laser Cutters, Lathes, Mills, Saws, Drills